

# SatQuery-Edge

Offline Satellite Intelligence Investigation Report  
SIH 26167 · ISRO · Prototype · Offline Mode

|                  |                            |
|------------------|----------------------------|
| Investigation ID | SQE-20260902-YG2XUP        |
| Timestamp        | 2026-09-02 18:12:43        |
| Intent           | single_image_vqa           |
| Tools Used       | vqa, grounding, geospatial |

## Query

describe the image

## Final Answer

Scene analysis complete.

=== Llama-3.2-1B-Instruct Edge SLM Analysis ===  
Scene contains extensive vegetation or agricultural fields, dense built-up structures, road networks or linear structures.  
Illumination appears normal.  
Approximately 796 distinct structural regions detected.  
  
Evidence Score: 82/100.

## Evidence Score

| Component            | Code | Value      | Weight |
|----------------------|------|------------|--------|
| Model Agreement      | M    | 0.880      | 0.30   |
| Sensor Reliability   | S    | 0.840      | 0.20   |
| Spatial Consistency  | P    | 0.887      | 0.20   |
| Temporal Consistency | T    | 0.820      | 0.15   |
| Input Quality        | I    | 0.598      | 0.15   |
|                      |      |            |        |
| FINAL SCORE          |      | 82.2 / 100 |        |

## Validation

Status: **PASS** ✓  
• 1 image(s) detected — count OK.  
• Image 1: Quality OK (mean=120, std=42.1).

## Detected Regions & Approximate Coordinates

| # | Label                         | Confidence | Area (km²) | Demo Georeferencing |
|---|-------------------------------|------------|------------|---------------------|
| 1 | candidate regions of interest | 0.86       | 138.471    | Yes                 |

## Specialist Tool Outputs

### VQA — SUCCESS — 3 region(s)

```
=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricultural fields, dense built-up structures,
road networks or linear structures.
Illumination appears normal.
Approximately 796 distinct structural regions detected.
```

### GROUNDING — SUCCESS — 1 region(s)

```
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'regions of
interest'.
Highest confidence: 0.86.
```

### GEOSPATIAL — SUCCESS — 0 region(s)

```
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 138.47 km².

■ Approximate demo georeferencing applied.
Demo center: (20.296°N, 85.824°E)
Coordinates are NOT derived from real GeoTIFF metadata.
```

## Execution Trace

```
[18:12:43] QUERY RECEIVED: describe the image
[18:12:43] ROUTER DECISION: Intent: single_image_vqa
Selected tools: vqa, grounding, geospatial
Images required: 1
[18:12:43] VALIDATION: Status: PASS
1 image(s) detected – count OK.
Image 1: Quality OK (mean=120, std=42.1).
[18:12:43] TOOL EXECUTING: vqa → started
[18:12:43] TOOL OUTPUT (vqa): Success – 3 region(s) detected
=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricultural fields, dense built-up structures,
road networks or linear structures.
Illumination appears normal.
[18:12:43] TOOL EXECUTING: grounding → started
[18:12:43] TOOL OUTPUT (grounding): Success – 1 region(s) detected
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'regions of
interest'.
Highest confidence: 0.86.
[18:12:43] TOOL EXECUTING: geospatial → started
[18:12:43] TOOL OUTPUT (geospatial): Success – 0 region(s) detected
Geospatial conversion complete.
Converted 1 region(s) to geographic coordinates.
Total approximate area: 138.47 km².
[18:12:43] EVIDENCE FUSION: Score: 82.2/100
M=0.880
S=0.840
P=0.887
[18:12:43] FINAL SCORE: 82/100
[18:12:43] ANSWER: Scene analysis complete.

=== Llama-3.2-1B-Instruct Edge SLM Analysis ===
Scene contains extensive vegetation or agricul
```

## Tool Versions

| Component               | Version                    |
|-------------------------|----------------------------|
| SatQuery-Edge Prototype | 1.0.0-SIH26167             |
| Python OpenCV           | 4.11.0                     |
| NumPy                   | 1.26.4                     |
| Streamlit               | 1.62.0                     |
| Report Engine           | ReportLab PDF              |
| Mode                    | OFFLINE — No external APIs |

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge